

Responsible Software - CheatSheet

IN BA5 - Cécile Hardebolle

Notes by Ali EL AZDI

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INTRO. Responsible Software. Software engineered in a responsible (ethical) way.

1. With a goal to do good. 2. While preventing avoidable negative impacts.

3. Taking people, other systems, social structures and our planet into account.

Active Responsibility. anticipating/preventing negative impacts before occurring.

Ethics. Principles/standards that guide human behavior in various contexts.

Metaethics. "what does it mean for an action to be right?"

Normative. "how should people act in general (moral principles)?"

Applied. (Our approach) "how can i act ethically in a real-world situation?"

Technical decisions can have *ethical* impli-**Ethical Decision Making.**

cations.

1. **Recognize**(knowledge+experience+habits)

Ethical Issue. any situation that may com-2. **Evaluate**(cognitive+reason+ffective) 3. **Act.**

promise respect of ethical value/principle. **Ethical Lenses.**

Ethical Agency. capacity to take actions that1. **Desirability.** product corresponds to what people want.

align with ethical values and principles. 2. **Viability.** product generates profits for a business.

Ethical Sensitivity. ability to 3. **Feasibility.** product can be created with existing or new

- recognize ethical issues (awareness) technology.

- identify their impact: who, what **Fourth criterion (Isaac et al.).** add **Ethicality.**

Ethical Dilemmas. decision between multiple alternatives, all conflict with ethical values/principles. *None of the options override the other, wrong-wrong choice.*

Moral side.

Emotions should be used explicite-ly/intentionally for responsible design (Roesser). In response to moral issues or motivating immoral behavior.

Trolley problem. thought experiment, **choice:** do nothing → runaway trolley kill people, or intervening → fewer people die. tests which moral principles justify action vs. inaction.

Ethical Decision Frameworks.

1. Identify the ethical issues

2. Get the facts

3. Evaluate alternative actions with *several ethical theories*

4. Choose an option and "test" it

5. Implement your decision and reflect on the outcome.

Stakeholder Individuals/groups/organizations/institutions/societies that might reasonably be affected by technology. **Direct Stakeholder.** *in contact* with the product

Indirect Stakeholder. affected through *consequences of the usage* of the software

Stakeholder analysis. Identifying every entity that can be affected by software positively/negatively. 1. Find all stakeholders possible using questions.

2. Use Direct vs Indirect Chart to find new stakeholders.

Additional Methodologies: *Ethix.* add *unintended stakeholders*, *Mitchell.* add *attributes to stakeholders*, *SAS.* use *different chart with scale for impact from product and influence on the product.*

Ethical Speculation. How to think, not what to think - anticipate potential impacts on stakeholders. Create a fictional story in two parts. (*done at the early design phase*)

I. Dark and Pessimistic

1. Imagine a character that would be negatively impacted by the software

2. Write a pitch that summarizes their story.

3. Find a title.

II. Positive outcome

1. Identify 1 or 2 main ethical issues represented in your story.

2. Describe the immediate and future consequences.

3. Imagine a happy ending for character.

SAFETY II. Levels of analysis. Macro (societies), Meso (groups/communities), Micro (individuals)

Internet Traffic. 99% submarine cables: 1.4M km, 4.7B users, 1.2 ZettaBytes (1.2 trillion GB).

Internet Benefits (+). Information access/dissemination, Communication, Social relations/organization, Identity/psychological development, Learning/cognitive development, Production/commerce.

Internet Harms (-). Privacy boundary violations, Social relationship harm, Community damage, Addiction, Cognitive impairment, Information overload, Knowledge/belief distortion, Democratic threats.

Tension. Same features enable both benefits/harms → *social connection enables social harm, information access causes overload, identity spaces blur privacy boundaries.*

From micro to macro.

Types of Impacts: lots of individuals, Social structures/institutions, Different for different groups.

Areas: Markets/economy, Knowledge/worldviews, health/well-being, Politics/citizenship, Environment.

△controversy Complex measurements, causality hard to establish.

Disinformation. false info, with intent to deceive.

Misinformation. false info. without intent to deceive

Malinformation. true info. used with harmful intent.

fake news → false information, with or without intent.

Cognitive effects. how we think.

Socio-affective effects. interaction with people and emotions.

System 1. fast/emotional/ activated first as quick response.

System 2. slow/logical/conscious analysis.

Illusory truth effect. repetition increases perceived truthfulness of claims by enhancing familiarity and intuitive/shortcut-based thinking.

Continued influence effect: misinformation continues to influence beliefs even after correction is received and accepted as true.

Increased use of digital media shows greater exposure to misinformation, increased political polarization, decreased trust, and increased hate. *empirical results remain debated and sometimes exaggerated*

Publication. publicly available content, *financial/political/coordinated* strategic incentives.

Human publishers. Influential figures/ordinary users publishing for visibility/revenue, often using *click-bait*/participating in organized campaigns (*coordinated troll operations*).

Social bots. Automated accounts publishing at scale (1–15%), they generate *disproportionately large share* of misinformation.

Propagation. content spreading through *reposting/liking/algorithmic amplification*.

Human propagation. content reshared by real/fake accounts. engagement inflated by paid workers, while moderation reduces resharing (*account bans lowering repost rates*).

Genuine resharing factors. Resharing driven by *inattention, emotional reactions, and socio-affective motives* such as signaling group membership or provoking reactions (*outrage*).

Social bots in propagation. Automated accounts rapidly *repost* and *target influential or opposing users* to maximize visibility and conflict.

Recommender systems. *select, rank, display* content, optimize for *engagement (popularity bias)*.

Ethics Canvas. Structure stakeholder, impact, and **Edge Cases.** Foresee consequences/identify opportunities for improvement systematically.

When to use? Early software development *Brain-When to Use: At all stages,design/major updates/user storm, visualize, mitigate ethical impacts?*

1. Identify Stakeholders: Who is affected?

2. Ethical Impacts: What are the impacts on them?

3. Actions: How to address these impacts?

Causal loop diagram. to explain complex system/s/improve performance/solve issues

focuses on specific issues (*amount of fake news, investment in an auto-detection system*)

Causal links are labeled with a +/– longer delays are a whole to reach a state/purpose; modeling simplifies represented with thicker arrows.

Cognitive side (Normative Ethics).

Utilitarianism(Consequences). choose the greatest good for greatest number

Deontology(Action). do right according to universal principles

Virtue(Person). act as a virtuous person

Care(Relationships). mutual responsibility/care for each other

choice: do nothing → runaway trolley kill people, or intervening → fewer people die. tests which moral principles justify action vs. inaction.

Example questions for brainstorming

- Who will use your product on a daily basis?

- Who is in geographical proximity with the users of your product?

- Who is in social proximity with the users of your product?

- Who is involved in the maintenance of your product?

- Who provides services that are needed to run your product?

- Who provides funding?

- Which communities are active in the application domain?

Indirect Environment Government Neighborhood

Publics) Direct End-users Friends

Admins Contractors Developers

SAFETY I. Security. Prevent the environment from affecting negatively the system. *△threat | Vulnerability scanning, ethical hacking, security audit, code review*

Safety. Prevent the system from affecting negatively the environment. *△hazard | Verification and testing, failure modes analysis, Redundancy*

Common challenge - human factor → consider software as **socio-technical system**

Internet/social media; human behavior/technical systems constantly influence each other.

Malfunction. software doesn't function/perform normally/as specified.

- Due to unintentional defect/fault/bug—critical consequences(△Life/Mission-critical systems)

Misuse/Abuse. attack, diversion, use of software for *nefarious purposes*.

Bad/threat actors. *malicious intent, authorized/unauthorized users*—direct/indirect impact.

Unintended use. use of software in a way not originally intended/not (officially) supported.

⊕ creates opportunities for innovation *use camera to scan documents* ⊖ use *llm to obtain factual information*

Intended use software used as intended. difficult to anticipate impact due to social △scale/time

User-Generated Content (UGC). any content posted by users online *images/videos/texts*

- Published publicly and (usually) not produced by professional/institution *wikipedia*

△Toxic content. material harmful/offensive can have negative impact on individuals/communities

Types of impacts? Physical/Financial/Emotional/Relational.

Hate and Harassment.

Violence threats, Identity misrepresentation

Self-inflicted harm. Eating disorder promotion

Ideological harm.

Extremism/terrorism, Misinformation

Exploitation. Sexual abuse, Scams

Hate speech

- Discriminatory/pejorative of individual/group

- calls out real/perceived "identity factors" of individu-*line/online*, subject to limitations that respect the rights of others and prohibit incitement to discrimination or violence, pillar of democracy.

- conveyed through any form of (images/memes/...) *UGC platforms provide channels for hate speech. Easily produced, shared, anonymous, global/diverse audience, realtime, amplified by content recommendation algorithms*

Central dilemma. protecting users from toxic content while preserving free speech rights

To decide?

- State (responsible for fulfilling rights),

- Platforms (private actors with business interests),

- People (responsible for the content they publish).

Free speech. right to express opinions and to seek, receive, share information/ideas, in any format of-*nation or violence, pillar of democracy.*

Content moderation:

False negative: toxic content that es-
capes moderation.

False positive: wrongful censorship,
restricting free speech.

TP.Model predicts positive and is correct

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$

Positive Predictive Value (PPV) = $\frac{TP}{TP+FP}$

Negative Predictive Value (NPV) = $\frac{TN}{TN+FN}$

False Discovery Rate (FDR) = $1 - PPV = \frac{FP}{TP+FP}$

False Omission Rate (FOR) = $1 - NPV = \frac{FN}{TN+FN}$

False Positive Rate (FPR) = $\frac{FP}{FP+TN}$

False Negative Rate (FNR) = $\frac{FN}{FN+TP}$

STRIDE Threat Modelling. Identify computer security—T: Tampering – Unauthorized modifications.

threats across six categories.

– R: Repudiation – Denial of actions without proof. – I: Information Disclosure – Data leaks.

When to Use: At any project stage, complementing bad actors modelling.

– D: Denial of Service – Service unavailability.

– S: Spoofing – Impersonation attacks.

– E: Elevation of Privilege – Gaining unauthorized access

Bad Actors Modelling Minimize vulnerabilities via risk assessment and mitigation.

When to Use: Design phase, prior to updates, audits, incident responses, and improvements.

What is a Bad Actor? A person/group exploiting a system for personal goals.

Motivations (might overlap):

- Money: Data theft, ransomware, sabotage by competitors.

- Politics: Misinformation/strategic sabotage/infrastructure attacks.

- Entertainment: Attacks on random systems, misinfo, spreading.

- Self-interest: harassing messages/commun. monitoring/location tracking.

- Ideas: Discrediting groups, data leaks, website defacement.

Steps: 1. Identify possible malicious users and their motivations.

2. Assess potential consequences for the system and users.

3. Develop measures to prevent these negative outcomes.

Harm Modelling anticipate/address potential harms users/stakeholders may encounter.

When to Use: At every stage of project, for every feature, until harm potentials are satisfactory.

Evaluate: 1. The category of use. 2. the types of harms. 3. the magnitude of harm

Categories of Use:

Harm from ...

Intended Use: correct usage.

Unintended Use: unexpected usage.

Misuse/Abuse: malicious use.

age.

Malfunction: system failures.

Types of Harm:

Humans: Physical, emotional injury.

Allocation of Resources: Opportunity, economic losses.

Human Rights: Dignity, liberty, privacy losses; environmental impact.

Social Systems: Manipulation, social detri-
ment.

Magnitude of Harm:

Severity: Impact on individual/group well-being.

Scale: How broadly the harm affects populations.

Probability: Likelihood of harm occurring.

Frequency: How often the harm might occur.

FAIRNESS I. Data in Software. people described by **attributes** (stored in databases/files), can be *sensitive*.
⇒ *ethical implications* such as *confidentiality, privacy, and fairness*. data is an *imperfect representation of real world*, must be *simplified* to fit context.

Data protection law (GDPR) - Personal data: reveals info about *identity*, directly/indirectly.

- **Special category** data: banned by default. *racial/ethnic origin, pol. opinions, rel. beliefs, health data.*

Nondiscrimination Law forbids direct/indirect discrimination, when **apparently neutral** practice puts *specific groups* at disadvantage. (*Protected Group/Class: Category qualifying for special protection by law/policy*)

Proxy Attributes. Variables that **correlate** w/ sensitive attributes.

Risks & Outcomes. can lead to **indirect discrimination** (*harder to detect*).

- Software has discriminatory outcomes even in absence of sensitive attributes in data. (*high school quality*) - correlates w/ income/ethnicity → becomes proxy.

Fairness. *Equality opportunities*; operationalized by treating people same. Scope matters: groups not protected law still face ethical harm.

Unfairness. Discrimination, stereotyping, decision errors based irrelevant factors; causing negative impacts via biased software/processes (*women get lower credit limits despite higher income*).

Bias. Systematic deviation from standard. context-dependent, not always harmful, possible ethical issues.

affects Problem definition/ideas/solutions/planning: shaped by external factors (info/org. culture)

1. **Pre-existing bias.** Bias already present world, captured in data (*systemic bias*).

2. **Representation bias.** Data inaccurately reflects target population/subgroup (*Sampling: over/under-representation. Population: sampled pop. differs actual users*).

3. **Measurement bias.** Methods yield inaccurate values, oversimplify reality, differ accuracy across groups (*blood-oxygen devices more accurate white people vs darker skin due melanin*).

4. **Cognitive bias.** Human (often implicit) deviations rational judgment; affects info processing/interpretation; enters via users/designers.

Stereotypes. Generalized beliefs about group (*implicit/explicit; pos/neg/neutral*); ignore individual differences, reinforced by design assumptions.

Group effects. groups with same opinions/preferences → reinforce confirmation bias; (*devil's advocate strategy*) doesn't work → groups with heterogeneous preferences more balanced

→ group-think: phenomena where alternatives are avoided, pseudo-consensus

Sunk-cost fallacy. continuing course of action just due to past investment → prevents alternatives→product with design flaws → **Technical debt.** future costs to fix, **Ethical debt.** ethical consequences/harms

Inclusive Design. design software that benefits the widest range of users by addressing diverse abilities, languages, cultures, and differences.

When to Use: Early in soft. development phases.

1. Identify needed capabilities for use.

2. Identify and label "Non-Average Users" (NAUs).

3. Add unaddressed capabilities/new user categories.

4. Develop and report concrete inclusive solutions.

Aim: Ensure artifacts match the needs of diverse individuals beyond average users.

Capabilities. perception(vision,hearing,...), thinking, action(dexterity,...)

Analyzing Values. Identify/analyze/evaluate values embedded in technology and resolve value-based tensions.

What are Values? principles/desirable goals for design, based on Schwartz's value theory.

1. Identify values/stakeholder values/artifacts values.

2. Analyze value-based benefits (supporting stakeholder values)/harms (opposing stakeholder values).

3. Map value tensions:

* Within the same value (e.g., benefit vs. harm).

* Between different values (interpret context to resolve).

FAIRNESS II. ML techniques inferring patterns from data, generalizable to unseen data. Process. Iterate to find parameters of function that best represent pattern in data. Result. Function + function parameters.	
Training. Run learning algorithm to find parameters → Inference. compute results on previously unseen data. Regression. Output continuous values (<i>house price from size</i>). Classification. Output discrete values (<i>wine vs beer from alcohol level, color</i>). <i>Supervised.</i> Output labels given. (<i>labels created by humans; picture descriptors</i>) <i>Unsupervised.</i> No output labels (<i>clustering, dimensionality reduction</i>). Data dependence. ML depends on data → sensitive to biases; in supervised, labels human-made → not all issues attributable to ML alone. Pipeline. Problem definition → data collect + preparation → model dev + model eval + model deploy. Model dev. Choose model type, objective function, hyper-parameter tuning for performance → unfairness risk when optimizing accuracy only. Aggregation bias. Combine groups with different patterns → model performs poorly for some groups, or mainly for majority. Evaluation bias. Benchmark not representative of served population → (<i>Gender Shades: intersectional bias across gender and race</i>). Deployment bias. Mismatch between intended problem and actual use → socio-technical bias. Fairness metrics. Mathematical measures of fairness notions (<i>criticised for oversimplification; focus outcome</i>). Error parity. Error rates independent of group. Contains: 1. Error rate balance. Equality of error rates compared to actual values→FPR, FNR across groups. 2. Conditional use accuracy equality. Equality of predictive values compared to predicted values→PPV,NPV across groups. 3. Equal accuracy. Equality of accuracy across groups. Demographic parity(Equal acceptance rate). Positive prediction rate independent of group (<i>focus outcome</i>): TP + FP Total Cannot satisfy all fairness metrics simultaneously. Group fairness criteria (GFC). Quantify equality between groups; bundle people treated differently → aim for group-level equality. → Mathematical impossibilities among parity measures, incl. with demographic parity. → Unfair world + imperfect classifiers → not all group fairness criteria hold simultaneously → choose priority notion of fairness. Individual fairness. Similar individuals should receive similar decisions; requires similarity metric, tricky. Causal analysis of fairness. Focus causes of unfairness and causal pathways from sensitive attributes to outcomes → find proxies via causal graphs. Improve fairness. Either modify data or modify model. Modifying the data. (<i>pre-processing</i>) If data biased, learned patterns biased. To reduce bias, <i>Fairness through unawareness</i> removes sensitive attributes from dataset, but reduces accuracy. If proxies exist, bias may persist. Other ways: weighting, resampling (ensure better representation of concerned groups), or transformation (transform data with function preserving data while hiding bias). Acting on the model. At training time: (<i>in-processing</i>) optimise accuracy and fairness together, add fairness metrics; or (<i>post-processing</i>) directly apply transformations to output, independent from model. To improve fairness of model, almost always face dilemmas: → Leveling down. More fairness for some might compromise fairness to others. → Fairness–accuracy. More fairness may often compromise accuracy. → Fairness–privacy. Access to sensitive attributes, increasing privacy may compromise fairness.	
Datasheet for dataset. Fill form with technical details/explanations about data, collection process, processing, intended usage→for devs: avoid publishing ill data; for users: ensure data cleanliness, better understand dataset. Model card. Same idea but for ML → technical info, intended use, performance, metrics, data for training/evaluation, quantitative analyses, ethical reviews, usage recommendations. AI Ethics card (IDEO). Guide ethically responsible, culturally considerate design with data. Principles: data ≠ truth. don't presume AI durability, respect privacy/collective good, unintended AI consequences → design opportunities. Goal: humanize data (<i>people behind data</i>). Method: find/read documentation, raw data, materials; select variables/columns; pick few; inspect their data; focus on one variable (select people random or by specific characteristics); write conclusion in terms of ethical impact and risks.	
Sustainability I. Capacity to endure across full lifespan (launch to end-of-life); linked to maintenance/evolution. Three pillars view. Economy, environment, society (treated as substitutable). Strong Sustainability. Recognizes the interdependence of economic, social, and environmental systems, emphasizing the irreplaceability of natural resources. Technical Sustainability. Focuses on software durability/its maintenance and evolution over time. Nested view. Economy part of society; society part of environment → not substitutable. Green by software. Software supporting sustainability goals of other systems. Green in software. Software itself designed to be sustainable (focus). When? - Development. Requirements, design, implementation, testing, distribution, deployment. - Usage. Operation, maintenance, evolution. - End of life. Decommissioning, disposal. What. Software cannot run alone → requires hardware and network together. Beyond carbon. Carbon footprint matters (<i>ICT sector: 1 × 10¹² kg CO₂, more than France or Australia</i>), but assessment should cover broader planetary boundaries (<i>9 critical processes</i>). Global warming. Increase in multi-decade global mean surface temperature above industrial level (1900): +1.1°C (2011–2020), +1.45°C (2023). due to human activity/greenhouse gases. Greenhouse gases (GHG). Atmospheric gases trapping heat in Earth's atmosphere (natural phenomenon); increased GHG concentrations raise temperature. Main contributors: CO ₂ , CH ₄ , N ₂ O. Software and electricity. Software needs electricity; electricity generation emits GHG → software has CO ₂ emissions depending on energy consumption and carbon intensity of electricity production. Carbon intensity examples. (<i>coal: 820 g CO₂/kWh</i>), (<i>gas: 490 g CO₂/kWh</i>), (<i>solar: 41 g CO₂/kWh</i>). Location. Electricity origin depends on location; carbon intensity computed from average energy mix (fossil fuels, nuclear, renewables). Time. Electricity mix varies over time; renewables not produced continuously. Optimizing software energy. Shift execution in time and location to exploit cleaner electricity mixes. GHG Protocol for companies. - Scope 1. Direct emissions from sources owned/controlled by company. - Scope 2. Indirect emissions from purchased/-plexity, loops, threads, data structures; also RAM, disk transfers, and I/O transfers matter. - Scope 3. Other indirect emissions across value chain (suppliers, customers/consumers). Programming languages and efficiency. Languages differ in efficiency; faster ≠ more energy-efficient. Main energy drivers. Majority from CPU: algorithmic complexity, loops, threads, data structures; also RAM, disk transfers, and I/O transfers matter. Hardware and OS role. Idle power draw exists; OS/hardware power-saving systems (low power modes) affect consumption.	
Data centers. Virtualization lets one physical server run multiple workloads; consolidation can reduce energy if optimized. Cooling critical; data centers powered continuously. Typical split: servers 50%, cooling 40%, remainder losses/other. Distributed software. Multiple components deployed across infrastructure → distributed energy consumption (including user devices) + network. Usage-stage split. 50% user devices, 25% data centers, 30% network. Network accounting. Estimation hard (constantly changing; boundary definitions vary). (<i>60 Wh/GB; phone charging: 22 Wh</i>) △Energy efficiency vs time, cost, performance, maintenance. Rebound effect (Jevons' paradox). Efficiency gains reduce service cost → increased consumption; existence widely accepted, magnitude debated. Direct rebound. Consumers use more of the same service. Indirect rebound. Savings spent on other energy-consuming goods/services.	
Decision Matrix. Make decisions by evaluating and prioritizing options based on multiple criteria. <i>When to Use:</i> To narrow down a list of options to one choice. When decisions must consider several criteria. 1. Brainstorm and identify evaluation criteria. 2. Refine criteria and assign relative weights (<i>dis-tribute 10 points</i>). 3. Create an L-shaped matrix with criteria and weights along one axis, and options along the other. 4. Rate each option consistently (<i>1–5 scale</i>). 5. Multiply weights by ratings, sum results, and select the option with the highest total score.	

SUSTAINABILITY II. FLOPS. Floating point operations → growing 4×/year since 2010. Examples of ML training footprint. GPT-3 (175B param) → 1300t CO ₂ e; BLOOM (176B param) → 433t CO ₂ e. Carbon budget. ≈ 2 t CO ₂ e/person/year if we want to keep global warming under control.	
PUE (Where). Data-center efficiency overhead (cooling + power delivery losses). CI (When). Grid carbon per kWh at that time (electricity mix). Training. training time in hours and power consumption in W and power usage. $C_{training} = t \times (P_{CPUs} + P_{GPUs} + P_{ME}) \times PUE \times CI$ <i>Energy per query:</i> BLOOM 0.004 kWh/query and 0.28 kWh in idle. Generative tasks most energy-intensive; image tasks > text (<i>can be as much as charging one phone</i>). Multi-purpose model > task specific model. $C_{ML} = C_{dev} + C_{inference}$ $C_{inference} \approx 60\%$ of total. C_{dev} can be large (one-time), $C_{inference}$ scales with usage. Embodied emissions. Greenhouse gas emission associated with production, transport, end of life → carbon already embedded into product (<i>36% embodied, 64% usage</i>). One-shot cost → attribute share based on exec time and used resources.	
Life Cycle Assessment. Methodology to evaluate range of environmental impact across all life cycle phases → impact on all, not just carbon footprint (<i>water, metals, ...</i>) → long process. Product Carbon Footprint (PCF). LCA reduced to Global Warming Potential and carbon. → Estimate with databases + math modeling. → Highest embodied emissions: integrated circuits (CPU, RAM, ...) + displays; mainly manufacturing; depends on component type, perf, dimensions. → Take embodied emissions, timeshare (<i>exec time / hardware lifespan</i>), resource share (<i>exec resources / face + groundwater</i>), incl. potable water (<i>most competition between technology and humans</i>).	
Green water footprint. Rainwater consumption (<i>plants/soil</i>). Gray water footprint. Water consumed to dilute pollutants so water meets quality standards. Energy tech footprints. Large footprint: biomass, hydropower. Low footprint: solar photovoltaics, wind. EU electricity (2015). Total water consumption for electricity production: 1.54 × 10 ¹² L. Thermal power plant. Heat-to-electricity; needs water ⇒ high consumption; depends on cooling type; similar issue in data centers; potable water can be used. Cooling. Sensitive to temperature/humidity; varies with climate, location, season, weather conditions → water footprint varies; considered indirect consumption. <i>In energy production, often a water vs. energy(carbon) tradeoff.</i> WUE Source. (<i>cooling direct, electricity indirect</i>). $WUE_{source} = \frac{\text{Annual site (cooling) water usage} + \text{Annual source-energy (electricity) water usage}}{\text{Energy of IT equipment}}$ Water cycle. Water changes state and location; never lost, but availability in time and place crucial.	
Ethical Decision Making. explore ethical aspects of decisions/evaluate options using multiple ethical lenses. <i>What is a Lens?</i> A normative ethical theory used to evaluate decision options. The 6 Lenses: – Rights: Respect stakeholders' rights. – Justice: Treat people fairly and equitably. – Which option ensures fairness? – Utilitarian: Maximize good, minimize harm. Which option benefits the most stakeholders? – Common Good: Serve the community, especially the vulnerable. Which option serves the community best? – Virtue: Strive to be an ideal person. What kind of person am I if I choose this? – Care Ethics: Prioritize relationships and feelings. Which option considers all stakeholders' concerns?	
EMPOWERMENT I. Choice architecture. Context/environment where decision is made. Nudge. Aspect of choice architecture altering people's behavior predictably without forbidding options or changing economic incentives → intentionally bias decisions. → Applied in gov, health care, promote greener behavior, info security, privacy. Social nudges. Reference to users of how other users behave → create social norm. Default. Choice assigned in absence of active deci- sion, without removing choice <i>auto subscription renewal, newsletters</i> → significant effect on chosen option. - Opt-in. Checkbox unticked → user actively ticks. - Opt-out. Checkbox ticked → user actively unticks <i>doubled organ donors(40%→80%).</i> → Threats to 3 aspects of people's autonomy. Factuality errors. Factually incorrect or fabricated content. Faithfulness errors. Content inconsistent with instruction- s/input, or without internal logic. Automation bias. More confidence in automated systems than other sources (including self) → over-rely on automated systems and recommendation systems even if output incorrect, even if own judgment would be correct. Dark patterns. Tricks in websites/apps that push unintended actions (<i>buying, signing up</i>). Design strategies that steer decision makers toward undesirable/sub-optimal options benefiting influencing party → outcome asymmetry, negative user impact. Some dark patterns modify user decision space; others modify information available. Evidence: > 40% of shopping-site timers reset or have no effect on advertised offer; fake social proof sometimes false. ≈ 11% of websites include deceptive patterns, including popular sites. Regulation. Deceptive patterns addressed in legal texts (<i>GDPR</i>); US FTC published “click-to-cancel” guidance (<i>one-click cancellation</i>).	
Attention-seeking patterns. Capture attention, reduce sense of agency, interfere with user control of technology → mental-health and social harms. Examples: <i>autoplay, pull-to-refresh, time fog, infinite scroll (social media addiction; originator Aza Raskin later apologized)</i> . Dilemma. Decision support often improves overall outcomes but introduces new errors vs non-supported decisions (+26% risk incorrect). Need account for automation bias and identify consequences (<i>false negatives, false positives</i>). Bias factors. Individual state; higher complexity → higher bias risk; time pressure; strongest driver: system design.	
The Digital Ethical Canvas. reduce ethical debt by systematically assessing ethical risks at all stages of the design/development process. <i>When to use it?</i> During the design and the development stages. Repeat the analysis several times throughout the engineering process. 1. Evaluate the benefits: List expected benefits 2. Evaluate the risks: Identify risks using the five ethical lenses, evaluate their probability and severity 3. Reduce the risks: For all risks, especially the highest, outline mitigation measures 4. Review: Assess if benefits outweigh risks and decide whether/how to proceed with the project 5. Repeat: Continuously reassess ethical risks	